Steffen Holter

HUMAN-AI COLLABORATION • RESPONSIBLE AI • AGENTIC SYSTEMS

sholter@ethz.ch | steffenholter.com

PROFILE

I study how AI systems can understand human intent and calibrate when to act, ask for clarification, or seek guidance. My research combines human–AI interaction, visual analytics, and reinforcement learning to design collaborative workflows with explicit, adjustable control. I develop conceptual frameworks and interactive systems, evaluated through user studies and agent benchmarks.

EDUCATION

ETH Zürich 2023 – Present
PhD in Computer Science
Thesis: Earning Initiative in Human-AI Collaboration

Imperial College London 2020 – 2021
MSc in Computing (Artificial Intelligence and Machine Learning)
Thesis: Human-in-the-Loop Explainable AI (Advisor: A. Aldo Faisal)

New York University 2014 – 2019
BSc in Electrical Engineering with a minor in Computer Science
Thesis: Variational Autoencoder Based Feature Extraction and Visualization

INDUSTRY

Adobe Research, San Jose, California, USA 2025
Research Scientist Intern (Advisors: Gromit Chan, Doga Dogan, Eunyee Koh)
- Developed persona-based browser agents and an interactive analysis workflow for usability evaluation of website prototypes.
- Connected aggregated usability findings to agent reasoning and interaction traces, enabling UX professionals to inspect issues, revise interfaces, and reassess changes.
- Research resulted in a full UIST 2026 paper, a patent application and an Adobe Sneak presentation.

Starship Technologies, Tallinn, Estonia 2021 – 2023
Senior Software Engineer, Autonomous Driving Team
- Responsibilities spanned the machine learning pipeline, from coordinating data annotation to training models and putting them into production on a worldwide fleet of autonomous delivery robots.
- Developed and deployed an end-to-end detection and segmentation model for road crossings, improving precision by approximately 8 percentage points at equivalent recall.

RESEARCH

ETH Zürich IVIA Lab, Zürich, Switzerland 2023 – Present
Doctoral Researcher (Advisor: Mennatallah El-Assady)
- Formulate frameworks to characterize human–AI collaboration across agency, interaction, and adaptation, guiding systems that support human control and shared responsibility in decision-making.
- Develop interactive analytics systems that infer user intent from interaction histories and identify ambiguity that requires clarification, helping users refine how their goals are interpreted.
- Investigate selective consultation in agentic workflows using reinforcement learning, with compact policies that balance task success against the cost of consulting stronger models or humans.

Imperial College Brain & Behavior Lab, London, UK 2021
Research Assistant (Advisor: A. Aldo Faisal)
- Used a reinforcement learning inspired approach to create a framework that learns customized user specific explanations based on individual behavior and interactions with AI systems.
- Formulated, designed, and conducted experiments introducing the human as an agent in AI assisted sequential decision making.

NYUAD Robotics Lab, Abu Dhabi, UAE 2019 – 2020
Research Fellow (Advisor: Anthony Tzes)
- Investigated area coverage strategies for collaborative drone swarms.
- Developed relative drone localization methods using 360° spherical cameras.
- Implemented computer vision algorithms to identify and track neighboring drones.

NYU VIDA Lab, New York City, USA 2018
Research Assistant (Advisor: Enrico Bertini)
- Developed visual analytics tools for interpreting and validating machine learning models.
- Designed and implemented a counterfactual explanation method supporting both local and global model interpretation.

PUBLICATIONS

UXCascade: Scalable Usability Testing with Simulated User Agents
Steffen Holter, Eunyee Koh, Doga Dogan, Gromit Chan
ACM Symposium on User Interface Software and Technology (UIST). 2026

Toward Agency in Human-AI Collaboration
Steffen Holter**,** Caterina Moruzzi, Mennatallah El-Assady
IEEE Computer Graphics and Applications. 2025.

Finding Needles in Document Haystacks: Augmenting Serendipitous Claim Retrieval Workflows
Moritz Dück*, Steffen Holter*, Robin Chan, Rita Sevastjanova, Mennatallah El-Assady
ACM Conference on Human Factors in Computing Systems (CHI). 2025.

Deconstructing Human-AI Collaboration: Agency, Interaction, and Adaptation
Steffen Holter, Mennatallah El-Assady
Computer Graphics Forum (EuroVis). 2024.

AdViCE: Aggregated Visual Counterfactual Explanations for Machine Learning Model Validation
Oscar Gomez*, Steffen Holter*, Jun Yuan and Enrico Bertini
IEEE International Conference on Visualization (VIS). 2021.

Relative Visual Localization for Unmanned Aerial Systems 🏆
Steffen Holter, Athanasios Tsoukalas, Nikolaos Evangeliou, Nikolaos Giakoumidis and Anthony Tzes
International Conference on Unmanned Aircraft Systems (ICUAS). 2021.

Development of a Versatile Modular Platform for Aerial Manipulators
Nikolaos Evangeliou, Athanasios Tsoukalas, Nikolaos Giakoumidis, Steffen Holter and Anthony Tzes
Service Robotics, IntechOpen, 2020

ViCE: Visual Counterfactual Explanations for Machine Learning Models
Oscar Gomez*, Steffen Holter*, Jun Yuan and Enrico Bertini
ACM Conference on Intelligent User Interfaces (IUI). 2020.

---- Preprints & Manuscripts Under Review ----

From Interactions to Intent: Classifying User Objectives from Provenance Logs
Steffen Holter, Tobias Stähle, Arpit Narechania, Mennatallah El-Assady
[Under Review]. 2026

Ask Your Advisor: Learning When to Consult Stronger Models in Agentic Systems
Pascal Sowietzki, Steffen Holter, Yannick Metz, Emil Kasper, Raphael Baur, Mennatallah El-Assady
[Under Review]. 2026

Let Me Clarify: Interactive Modeling of Soft Attributes
Steffen Holter, Rita Sevastjanova, Michaela Benk, Mennatallah El-Assady
[Under Review]. 2026

From Confidence to Contribution: How AI Peers Shape Metacognitive Processes in Groups
Antonia Schlieder, Steffen Holter, Jan Rummel, Filip Sadlo
[Under Review]. 2026

PostHocer: Method-Agnostic Global Correction for Dimensionality Reduction
Erges Mema, Steffen Holter, Hyeon Jeon, Mennatallah El-Assady, Rita Sevastjanova
[Under Review]. 2026

SKILLS

Python, PyTorch, C++, TypeScript, React, LLM Agents, Empirical Evaluation, Reinforcement Learning

TEACHING

Fundamentals of Web Engineering, Teaching Assistant 2024, 2025, 2026
Interactive Machine Learning, Teaching Assistant 2024, 2025, 2026

SERVICE

Reviewer: CHI 2027, UIST 2026, DIS 2026, VIS Posters 2026, DIS 2025, CHI LBW 2025

HONOURS AND AWARDS

ICUAS 2021 Best Paper Award 2021
Microsoft Foreign Studies Master´s Scholarship 2020
Post-Graduate Research Fellowship 2019
Phi Beta Kappa National Honour Society 2019
New York University Honour Scholar 2019
FICO Explainable Machine Learning competition, *Honorable Mention* 2018

MENTORING

2026 Pascal Sowietzki, MSc Thesis
Adaptive Agency: A Reinforcement Learning Framework for Agentic AI Systems

2026 Marie-Louise Dugua, MSc Thesis
User-Conditioned Modeling of Subjective Attributes for Personalized Recommendation

2026 Erges Mema, MSc Thesis
Interactive Neighborhood Correction in Projection Algorithms

2025 Alişer Ulaş Göner, BSc Thesis
Interactive Clarification for Evolving Attribute Spaces in Multi-Criteria Text Ranking

2025 Carlo Simoni, BSc Thesis
GraphWise: An Interactive, AI-assisted Platform for Learning Graph Algorithms

2024 Maria Gkoulta, Semester project
BioFitViz: Interactive Biometric Data Analysis